



EXAMEN PARCIAL DE CALCULO INTEGRAL

1- Si $I_n = \int x^n (a^2 - x^2)^{3/2} dx = -\frac{x^{n-1}(a^2 - x^2)^{3/2}}{n+2} + k a^2 I_{n-2}$
Hallar k (4 puntos)

2-

a) Integrar $\int \frac{\cosh x \, dx}{\sinh^5 x \sqrt{\sinh^2 x - 1}}$ (3 puntos)

b) Evaluar $\int_{\ln 1}^{\ln \sqrt{3}} e^{-x} \arctg(e^x) \, dx$ (3 puntos)

c) Evaluar $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\sec x}}{\sqrt{\sec x} + \sqrt{\csc x}} \, dx$ (3 puntos)

3- Demostrar que $\frac{\pi}{3\sqrt{2}} \leq \int_0^{\frac{\pi}{3}} \frac{1}{\sqrt{2 - \sin^2 x}} \, dx \leq \frac{2\pi}{3\sqrt{5}}$ (3 puntos)

4- Si $\int_{x g(x)}^x f(t) \, dt = \int_0^x \sqrt{1 - (f'(t))^2} \, dt$ y $g(x) = \int_x^{2x} \frac{\sin(\frac{t}{x})}{t^2} \, dt$
 $f(0) = 0$

Calcule $\int_0^{\pi} 2(f(x))^2 \, dx$ (4 puntos)